Ibrahim Alabdulmohsin

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Executive Summary

I have a passion for deconstructing complex problems and developing solutions rooted in first principles. Currently, my research explores pushing the boundaries of parameter efficiency through techniques such as scaling laws, shape optimization, and recursive inference, without compromising on critical dimensions like fairness, safety, and cultural diversity. I am particularly excited about applying such techniques to teach foundation models to solve complex combinatorial problems.

Professional Experience

Google DeepMind — *Nov 2019 – Present*

Research Scientist | Zürich, Switzerland

- **AI Scaling & Efficiency:** Leading research on parameter efficiency, including pioneering shape optimization via scaling laws (SoViT), layerwise reinitialization, and recursive inference to enable "System 2" latent thinking. Contributed to major efforts for vision/multimodal systems (FlexiViT, NaViT, PaLI, Gemma, PaliGemma, SigLIP2, ...).
- **Ethics & Robustness:** Led initiatives to ensure models are fair and robust across cultural and socioeconomic dimensions, including projects on mitigating biases, improving cultural diversity, and red-teaming.

Saudi Aramco — *Nov 2017 – Oct 2019*

Head of Enterprise Analytics & Technical Lead of Digital Transformation | Dhahran, Saudi Arabia

- **Digital Transformation Leadership:** Founded and led the company's first Advanced Analytics team. Served as Technical Lead for the corporate Digital Transformation Program, accelerating AI adoption across the world's largest energy infrastructure.
- **Process Optimization:** Oversaw the design and deployment of ML applications for downstream, supply chain, sales, HR, finance, and IT, among other sectors.
- **Talent Ecosystem Building:** Launched the "Data Science" specialist development program.

Education

King Abdullah University of Science and Technology (KAUST) — *2012 – 2017*

Ph.D. in Computer Science | GPA: 4.0/4.0

- **Thesis:** "Learning via Query Synthesis". Committee: X. Zhang (KAUST), X. Gao (KAUST), D. Keyes (KAUST), W. Wang (UCLA).
- **Foucs:** Theoretical machine learning; e.g. active learning, query synthesis, statistical learning theory and information theory.

Stanford University — *2009 – 2011*

M.S. in Electrical Engineering | GPA: 4.15/4.0

- **Focus:** Networked Systems, Artificial Intelligence, and Operations Management.

University of Nebraska-Lincoln — *2000 – 2005*

B.S. in Computer Engineering, **Minor in Economics** | GPA: 3.98/4.0

- **Honors:** Highest Distinction; Superior Scholarship Award.

Community Leadership & Global Outreach

- **Global Policy:** Member of the **2019 S20 Task Force (Digital Revolution)**, serving as a scientific advisor to the G20 on digital governance. Also, chair of of the AI track at the **2019 Arab-American Frontiers**

Symposium (organized by the U.S. National Academies), held in Cairo, 2019.

- **Regional Development:** Co-organizer of **MenaML** (Middle East & North Africa Machine Learning) Winter School (`https://mena.ml`). Our mission is to promote the development of AI talent in the region.
- **Academic Stewardship: Area Chair** for *NeurIPS* and *ICML*. **Senior Area Chair** at *NeurIPS Datasets & Benchmarks* track. Recipient of "Outstanding Reviewer" awards at *ICML* 2020 and *NeurIPS* 2016, 2022, 2023.
- **Media Coverage:** Was profiled on the Saudi National TV program **"Josoor" (Bridges in Arabic)**, a 30-minute documentary highlighting my academic and professional journey.

Books

During the doctoral study, authored the book *Summability Calculus: A Comprehensive Theory of Fractional Finite Sums*, published by Springer in 2018. Many results were later featured in popular websites, e.g.:

- **Wikipedia:** Closed-form expressions relating the Euler-Mascheroni constant to approximation errors (see Euler Constant) and identities involving Gregory coefficients (see Gregory Coefficient).
- **OEIS:** Sequences and series summation results archived in the *On-Line Encyclopedia of Integer Sequences* (see A002206).